

Le système international d'unités The International System of Units

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Annexe 2

Annex 2

Valeurs recommandées des fréquences étalons
destinées à la mise en pratique de la définition du
mètre et aux représentations secondaires de la
seconde

**Recommended values of standard frequencies
for applications including the practical
realization of the metre and secondary
representations of the second**

Partie 2

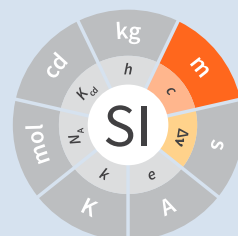
Part 2

Ytterbium 171 Ion ()

Ytterbium 171 Ion ()

CCL-CCTF Frequency Standards Working Group

May 2016
mai 2016



**Recommended values of standard
frequencies for applications including
the practical realization of the metre and
secondary representations of the second
Part 2: Ytterbium 171 Ion ()**

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1. Recommended value [1] of the frequency

$$f(^{171}\text{Yb}^+, \text{ octupole}) = 642\,121\,496\,772\,645.0 \text{ Hz}$$

equivalent to

$$\lambda(^{171}\text{Yb}^+) = 466\,878\,090.060\,497 \text{ fm}$$

with an estimated relative standard uncertainty of 6×10^{-16} .

This radiation is already endorsed by the CIPM as a secondary representation of the definition of the second.

2. Source data

Adopted value $f(^{171}\text{Yb}^+, \text{octupole}) = 642\,121\,496\,772\,645.0 \text{ Hz}$
 $u_c/y = 6 \times 10^{-16}$

calculated from

$f(^{171}\text{Yb}^+) / \text{Hz}$	u / Hz	$f(^{171}\text{Yb}^+ \text{ oct}) / f(^{171}\text{Yb}^+ \text{ quad})$	u	source data
642 121 496 772 657.0	12			[2]
642 121 496 772 645.15	0.52			[3]
642 121 496 772 646.22	0.67			[4]
642 121 496 772 644.91	0.37			[5]
642 121 496 772 645.36	0.25			[6]
		0.932 829 404 530 964 65	3.1×10^{-16}	[5]

as a weighted mean from the above values applying the method of ref. [7]. Considering that the values came from only two laboratories, the CCTF considered it prudent to enlarge the uncertainty by a factor of two.

^{171}Yb ion, $^2\text{S}_{1/2}-^2\text{F}_{7/2}$ unperturbed optical transition

References

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<http://www.bipm.org/jsp/en/CIPMRecommendations.jsp>
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